

JOSHUA CARBON

COMPUTER ENGINEER / SOFTWARE DEVELOPER

Software Development • Web • AI-Enabled Systems • Integration

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SUMMARY

Computer Engineering graduate with hands-on experience building and integrating software across web, desktop, and AI-enabled systems. Delivered a real client platform (AIQUANTA) spanning frontend development, API/HTTP integration, and AI-assisted functionality; built a Microsoft Copilot Studio AI assistant (CompetencyIQ); and developed a custom-trained YOLOv5 computer-vision system. Comfortable working across the stack — from OOP desktop applications to responsive web frontends — with a consistent focus on debugging, system integration, and clear technical documentation.

SKILLS

Programming: C#, Python, C++, C, TypeScript, Visual Basic

Web Development: React, TypeScript, Vite, React Router, HTML, CSS, REST/API Integration

AI & Automation: YOLOv5, OpenCV, OCR, Roboflow, Google Colab, Microsoft Copilot Studio, Prompt Engineering

Software Development: OOP, Windows Forms, Git, GitHub, Debugging, Software Testing

Tools: Microsoft Visual Studio, Visual Studio Code, Postman, Microsoft Office

Embedded / IoT: Arduino, Raspberry Pi, ESP32/ESP8266, Sensor Integration, Embedded C

PROJECTS

AIQUANTA Training Consultancy Services — Paid freelance/client project. Implemented a client-facing consulting website and operations platform using React, TypeScript, and Vite, translating client requirements into a responsive, component-based frontend. Integrated a Cloudflare Worker HTTP API with D1 relational storage for data handling, built authenticated administration and reporting/analytics features, and added AI-assisted inquiry-relevance functionality alongside Microsoft Graph email integration. Handled deployment, technical SEO, and security considerations, refining functionality through iterative client-driven development and debugging.

CompetencyIQ AI — Built an AI assistant using Microsoft Copilot Studio and prompt engineering to support workforce competency intelligence, structuring its behavior around employee profiles, training history, post-training assessments, technical certifications and expiry, training-needs analysis, audit/quality findings, and operational risk indicators.

Personal Portfolio Website — Designed and built a React, TypeScript, and Vite portfolio site with React Router-based project case-study pages, a role-tailored resume selector, responsive/adaptive layouts, and technical visualizations; deployed to Netlify with iterative debugging and refinement.

MotorPH Payroll Management System — Developed a desktop payroll system in C# and Windows Forms covering employee records, attendance, payroll computation, leave management, and reporting.

QA Automation Framework — Built a reusable web automation framework using Python, Robot Framework, Selenium, and SeleniumLibrary, with reusable automation components, automated test execution, and reporting.

AI-Based Japanese Character Recognition using YOLOv5 — Trained a custom YOLOv5 model (Roboflow, Google Colab) and built a Python/OpenCV OCR pipeline on Raspberry Pi for real-time Kanji recognition — 95.33% accuracy across 50 Kanji classes and 150 test trials.

EXPERIENCE

Intern / OJT — Erovoutika Robotics and Automation Solution | May 2023 – Aug 2023

- Worked alongside engineering teams to troubleshoot technical issues across automation systems and embedded components.
- Performed debugging and calibration on automation units, supporting consistent performance during operation.
- Supported integration of sensors and embedded components into robotics and automation systems, including research/development work on a line-following robot.
- Authored technical documentation covering setup, calibration, and troubleshooting procedures for engineering reference.

EDUCATION

Bachelor of Science, Computer Engineering — Mapúa University | 2019 – 2025

Bachelor of Science, Electrical Engineering — Mapúa University | 2014 – 2019